BWSI Medlytics Challenge 1: Predicting Hypothyroidism

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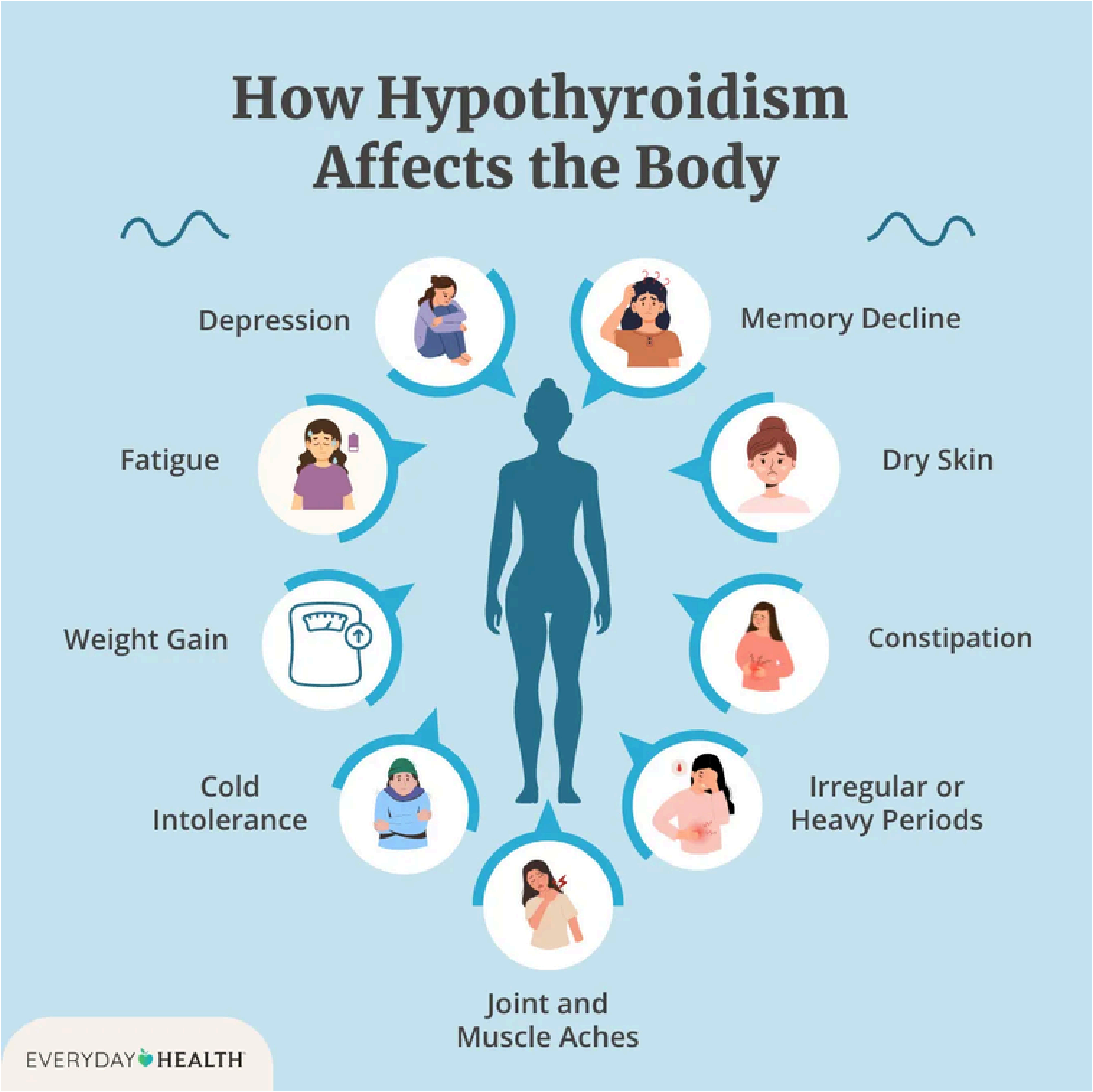
Problem/Motivation

- Hypothyroidism is a common endocrine condition in which the thyroid gland doesn’t produce enough thyroid hormones.
- Low thyroid hormones often lead to issues such as fatigue, weight gain, sensitivity to cold, cardiovascular issues, and developmental problems.
- If left untreated, hypothyroidism can lead to significant morbidity and mortality.
- In the United States, an estimated 20 million Americans have some thyroid disease, and up to 60% of these individuals are unaware of their condition.



Problem/Motivation

- Healthcare providers are in need of reliable methods to identify patients who have hypothyroidism.
- Early and accurate detection allows healthcare professionals to greatly improve patient outcomes.
- The collection of complete patient data is often an expensive and time-consuming process.

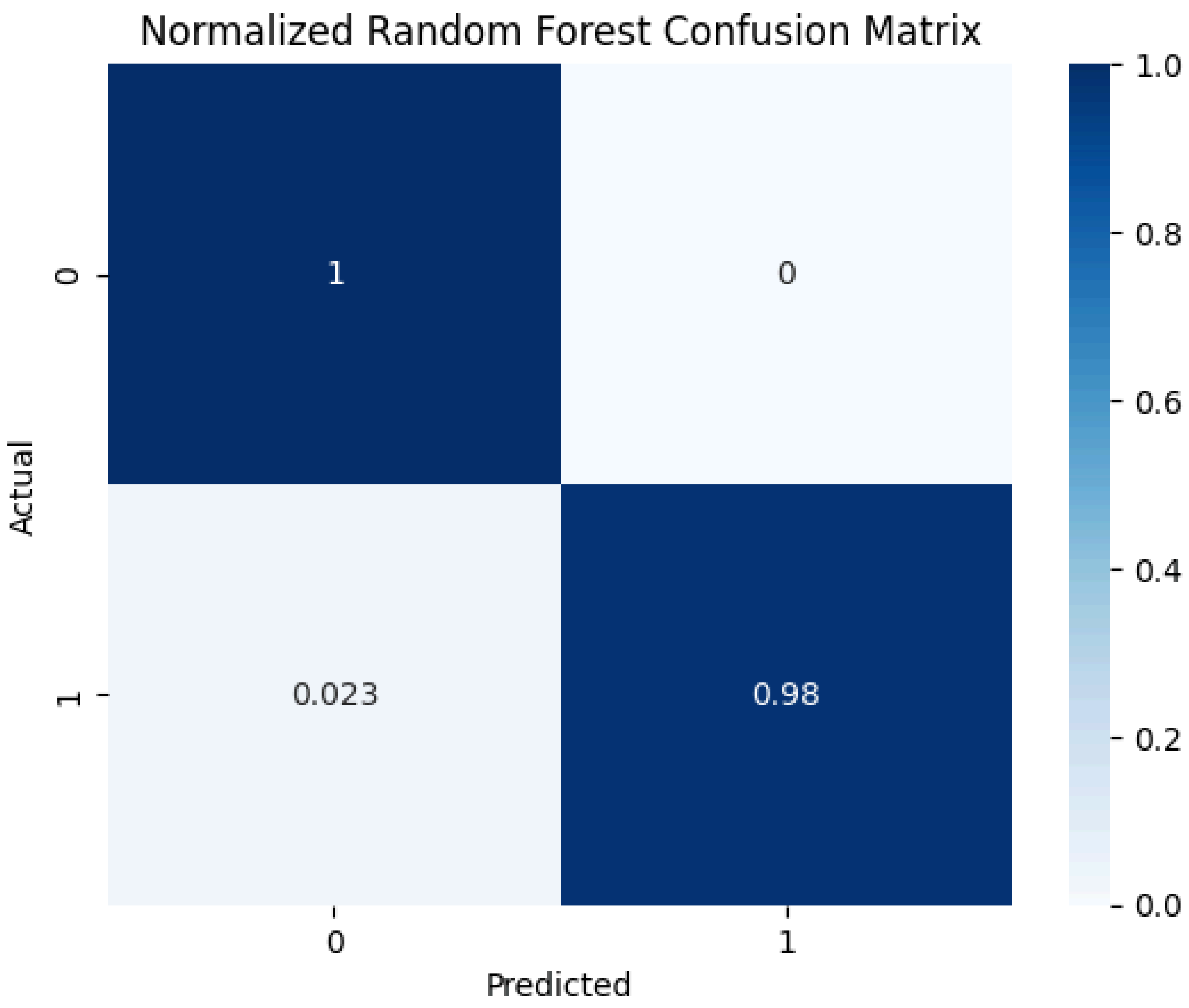
Research Question

Can we build a model that accurately predicts hypothyroidism while using the smallest possible number of important patient features?

Methodology

1. **Load allhypo csv files**
   2800 train, 972 test, ? become NaN
2. **Binarize Class Column**
   strip |ID, hypothyroid becomes 1
3. **Stratified 80/20 Split**
   2240 Train, 560 Val, 44 Val Pos
4. **Data Preprocessing**
   Drop Labels, Fill NaN with median for numeric and mode for categoric.
5. **Train Model**
   SVM/DecisionTree/RandomForest/LogisticRegression
6. **Score on Val**
   AUC and Confusion Matrix Metrics

Results

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Confusion Matrix Metric:  1.9318181818181819
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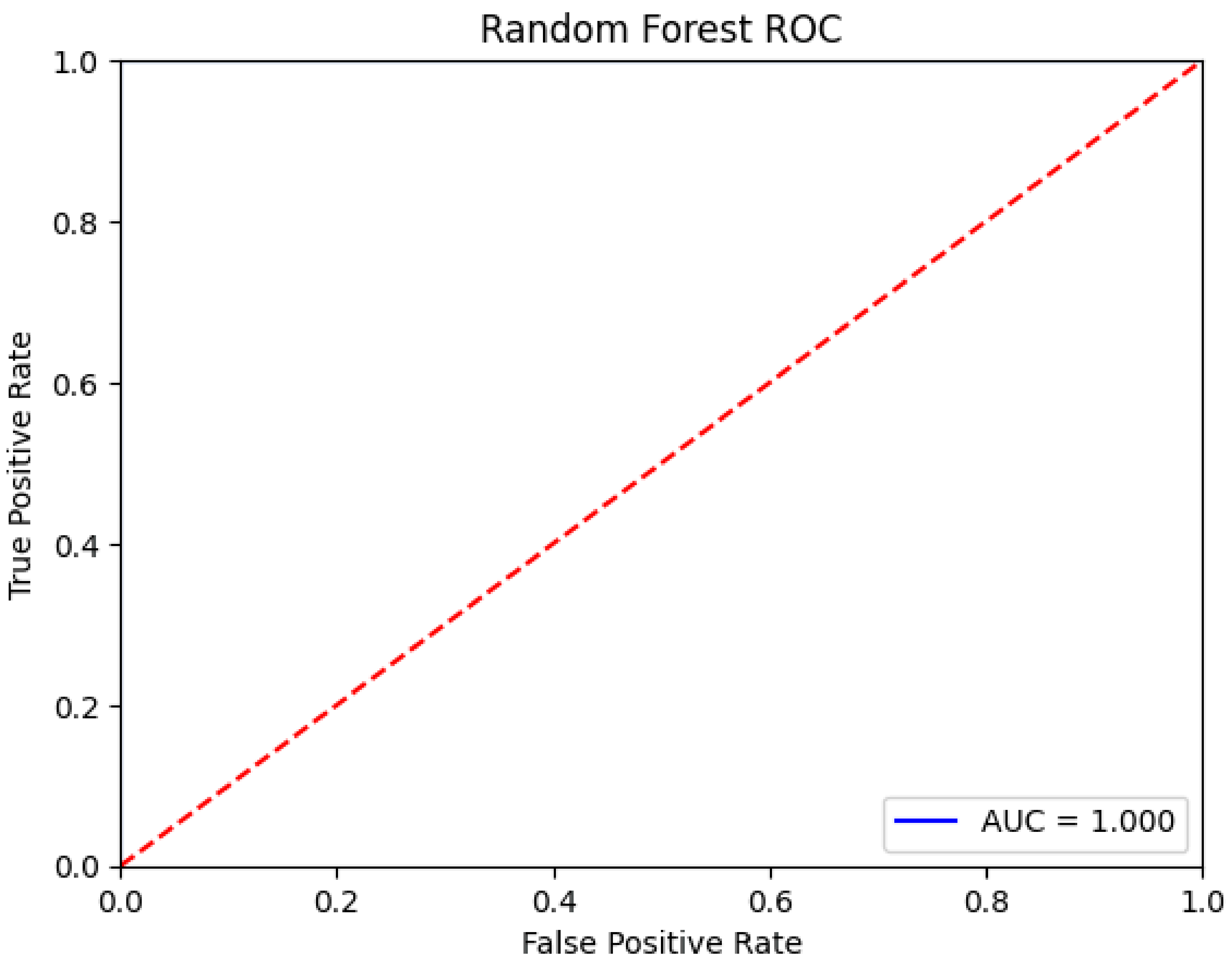
Interpretation

These results show that our Random Forest model performed extremely well at identifying healthy patients (nearly 0). It also classified nearly 100% of non-hypothyroid cases, with very few false positives, meaning the model rarely mislabeled healthy individuals as hypothyroid.

For patients with hypothyroidism, the model identified 98% of the cases, missing only 2.3%. This indicates that our model has very high sensitivity.

Results

AUC: 1.0



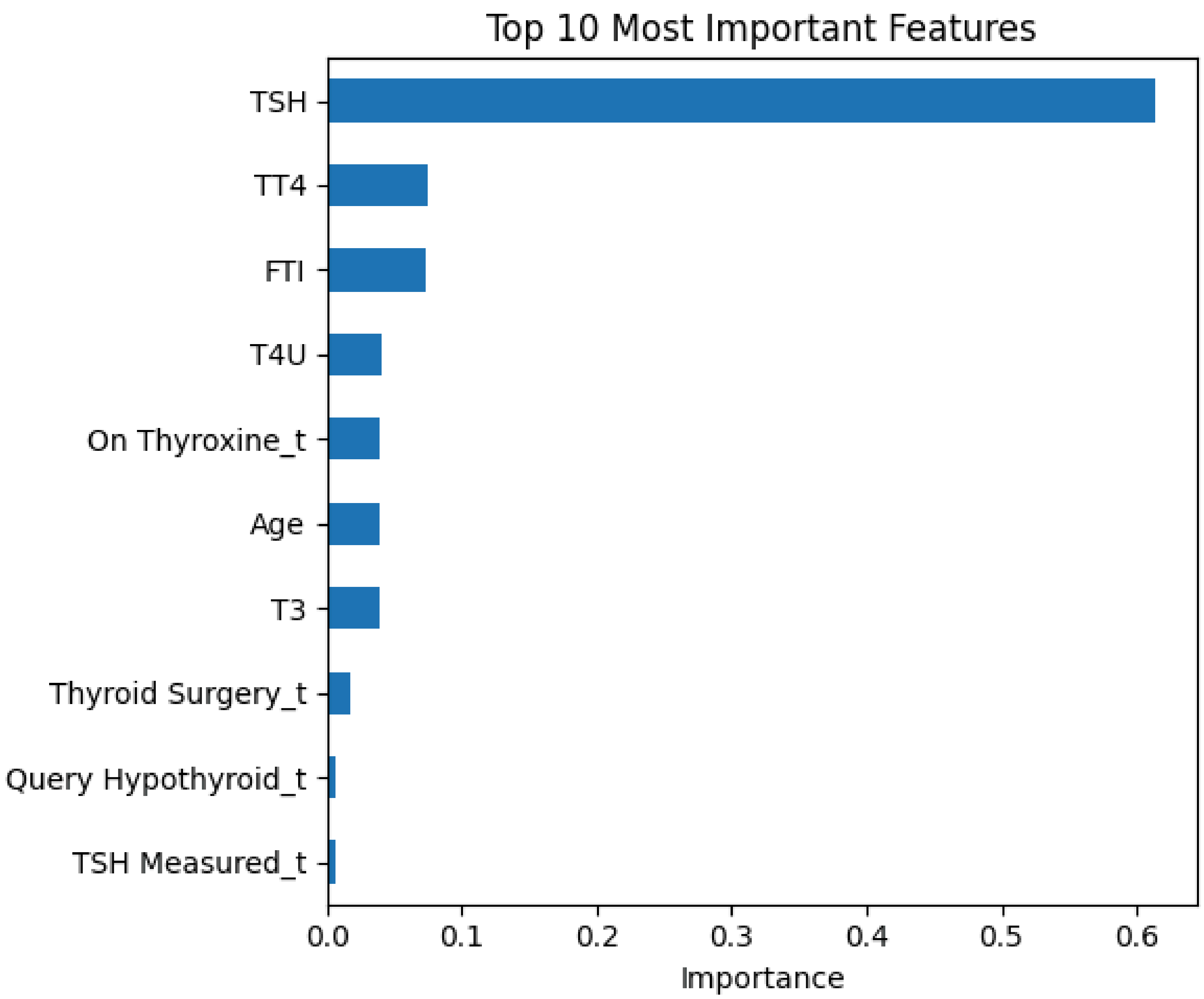
Model Score on Validation: 0.9982142857142857

Interpretation

Our model perfectly distinguishes between positive and negative cases in our validation dataset. An AUC of 1 means that the model doesn't get any false positives.

Though this was surprising, we made sure we had no data leakage, and we used min_samples_leaf=5 to try to prevent overfitting

Results



Interpretation

Our results show that TSH (Thyroid Stimulating Hormone) is by far the most important feature for predicting hypothyroidism. TSH has an importance score of around 0.61. The following most important features are TT4 and FTI; however, their importance scores are much lower compared to TSH, which suggests that they have a much lower impact.

These results are expected because TSH is one of the main lab tests used by doctors to diagnose thyroid disorders. Other features such as TT4. FTI and age are a lot less directly related to thyroid issues, so they're less important in the model.

Limitations

- Small Dataset (specifically positives)
  - Imbalance in testing data - 7% with positive diagnosis.
- Data Quality
  - A significant amount of missing or mislabeled data (Individual with age 455?) (110 without sex values)
    - Forced to use SimpleInputer (Imperfect solution)
      - For Numeric Columns, missing entries were assigned the median value
        - Issues - I.E missing TSH values default to ~1.2 (Normal)
      - For Categoric Columns, missing entries were assigned the mode value
- Missing TSH test impact
  - Patients with missing TSH tests (369) had “0 cases” of hypothyroidism
    - It can be inferred that because they did not get tested, they cannot be diagnosed with it. They very well could have the condition, however.
    - This is a limitation in the data, and the model fills in the missing value with ~1.2 (corresponding to a negative diagnosis). Thus, it arrives at the “correct” answer of negative, despite these cases in reality not being “negative” as much as “unknown” (due to lack of testing).
    - Given the 369 cases of such, using the 7% rate present in the rest of the data, this can be implied to be roughly 29 missing positives in the dataset that the model is labelling as negative.
- Binary Yes/No for Hypothyroid - does not predict specific types (need more data)

Conclusion

Our Random Forest model successfully predicted hypothyroidism with a **99.82%** accuracy using a small set of informative patient features.

Three key takeaways:

1.

TSH was the most important feature

2.

TT4, **FTI**, and **T4U** were the next most useful lab values.

3.

The model stayed accurate with small feature set, supporting **faster** and **cheaper** data collection

Thanks!